

Session Objectives

- Discussion of hypotonia and other neurologic findings in infants and children- presentation and initial management
- Discuss various conditions with hypotonia as a primary feature and their clinical features and presentation
- Understand clinical significance of various conditions causing abnormal head shapes in infants and children
- Discuss clinical aspects related to infection
- Discuss neurologic issues associated with trauma

Cerebral Palsy

- **Non-progressive** motor impairment due to damage/ dysfunction of the brain or spinal tracts
- Different types- Spastic, ataxic, athetoid, mixed
- **Spastic** most common- **upper motor neuron lesion- increased deep tendon reflexes, spasticity, sustained clonus, positive babinski**
- Birth history/ prematurity/ hypoxic event
- **Hypotonic at birth and newborn period** and progressively hypertonic and hyper-reflexia

Spinal Muscular Atrophy

- **Type 1:Wernig-Hoffman Disease-**
severe infantile form
- Type 2: late infant and slowly progressive
- Type 3
- Type 0- fatal perinatal
- **Severe hypotonia**
- Thin muscle mass
- Absent DTR
- Resp distress, poor feeding
- **Tongue Fasciculations**
- Milder- wheelchair, severely handicapped
- **Normal Intelligence**

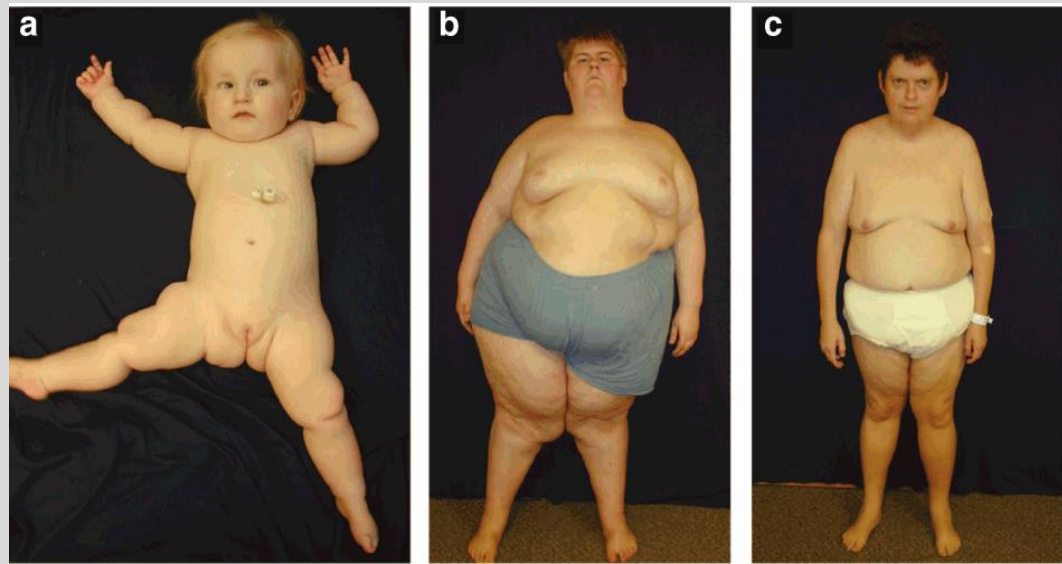
Down's Syndrome

- Common physical findings: flattened occiput, epicanthal folds, midface hypoplasia, upslanting palpebral fissures, single palmar crease, widened gap between big and second toe, enlarged tongue
- Higher incidence of congenital heart disease, anomalies of GI tract (duodenal atresia), leukemia, visual and hearing impairments
- **Generalized hypotonia**
- Joint laxity



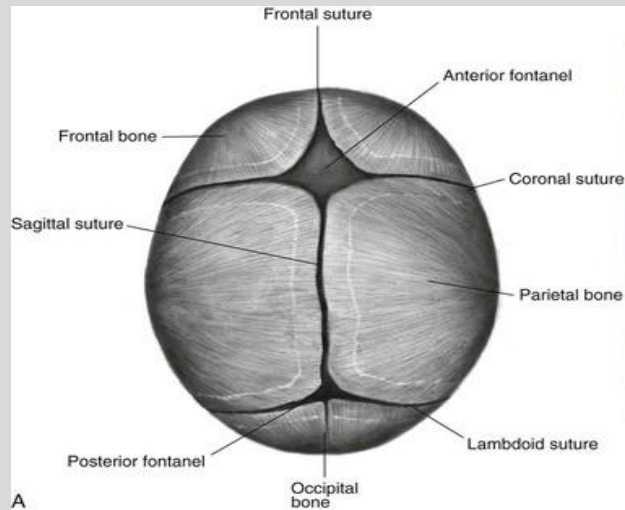
Prader-Willi Syndrome

- Deletion of paternal Chromosome 15q11.2-q13)
- Common findings: mild-moderate intellectual impairment, extreme hyperphagia, obesity, short stature, hypogonadism, poor sucking, small hands and feet
- **Infants with hypotonia and Failure to thrive, as child becomes hyperphagia and obesity** and all related complications- need extreme diet measures to control weight



Baby Head...What's normal

- Anterior Fontanel (6-18months close)
- Posterior Fontanel (6-8weeks close)
- Sutures
- Molding



Head Shapes...still cute!

- Micro Premie common **dolichocephaly**
- **Brachycephaly**- short, wide head, can be positional or coronal suture premature fusion
- **Positional plagiocephaly very common**
 - More belly time when awake, ROM neck exercises if torticollis, Helmet sometimes
- Still need to sleep on back for SIDS



Normal



Plagiocephaly



Brachycephaly



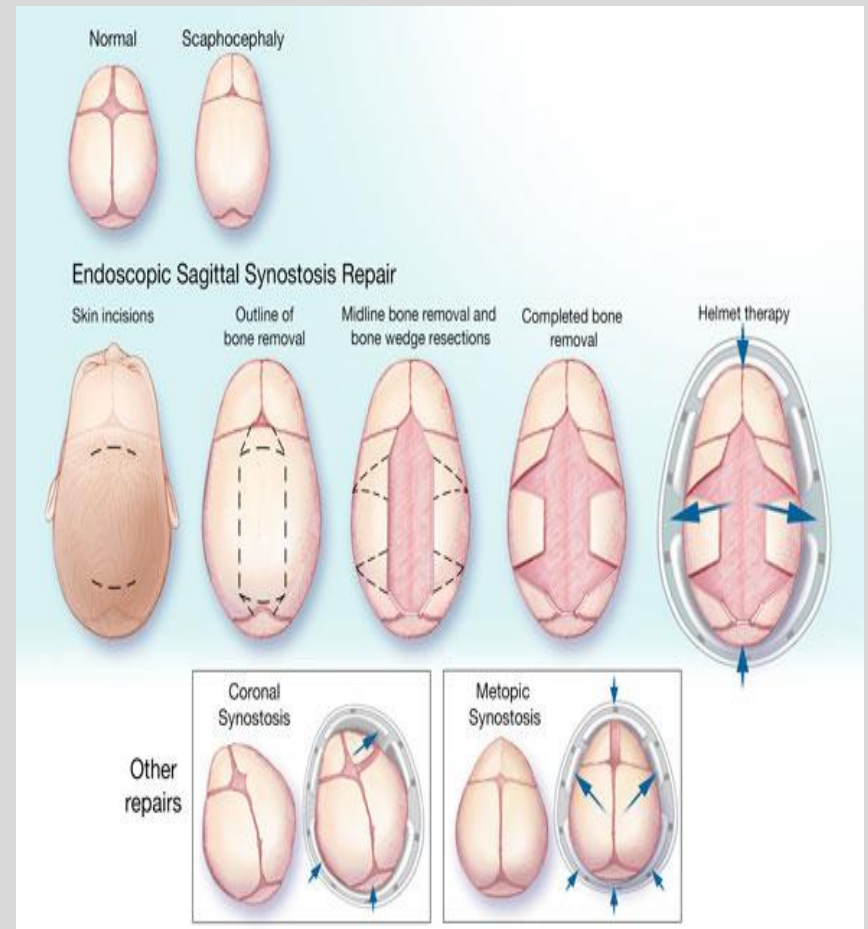
Dolichocephaly

Treatment

Helmets

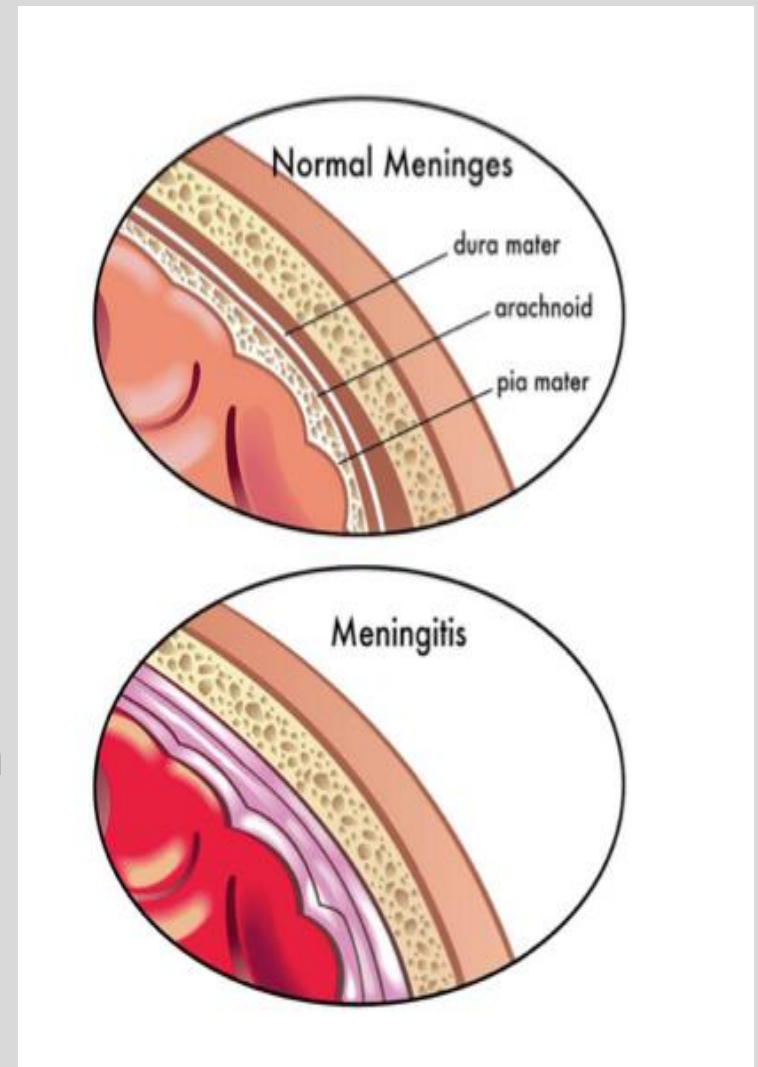


Surgical



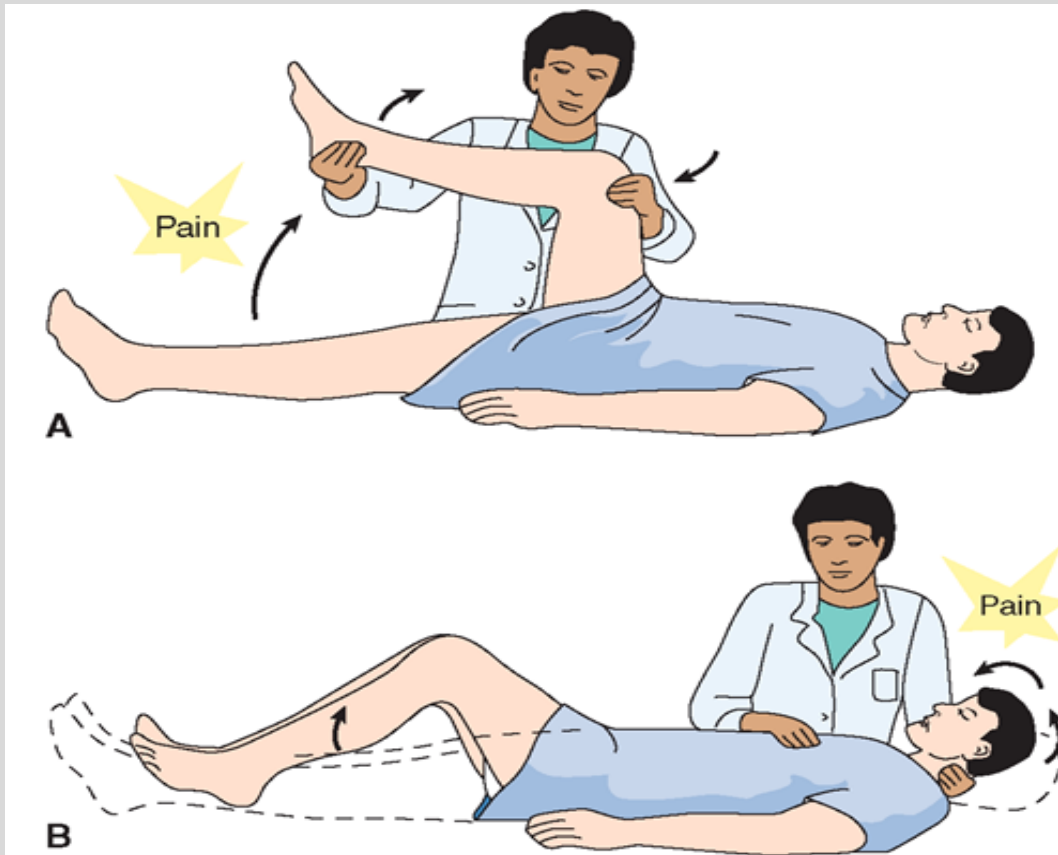
Meningitis

- **Symptoms:** headache, nausea, vomiting, anorexia, restlessness, altered state of consciousness, irritability
- **Signs of CNS infection:** fever or hypothermia, photophobia, neck pain, rigidity, obtundation, stupor, coma, seizures, focal neurologic defects
- **Signs of increased ICP:** headache, emesis, **bulging fontanel**, widening of sutures, oculomotor or abducens nerve palsies, **apnea**, **hyperventilation**
- **Cushing's triad** (Irregular respirations, Hypertension, Bradycardia)



Meningitis Physical Exam Signs

Not reliable in infants, young children!

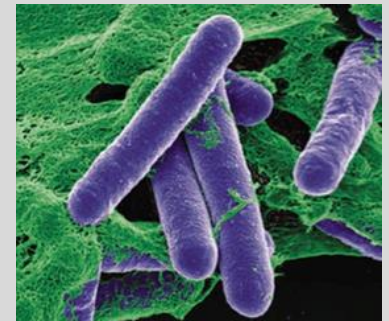


Kernig's sign

Brudzinski's sign

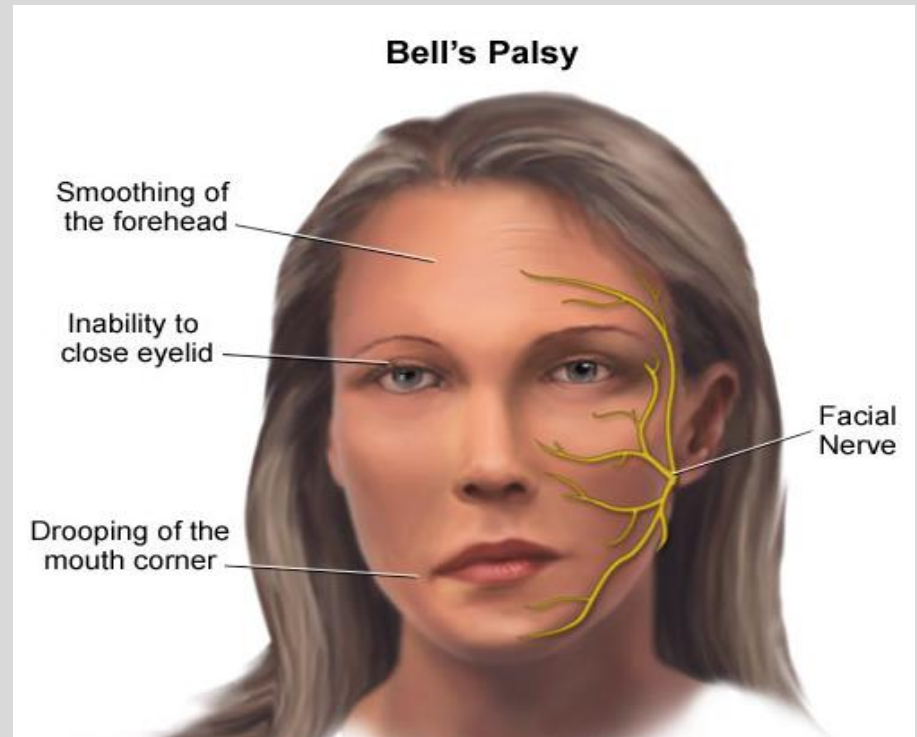
Infantile Botulism

- *Clostridium botulinum*: anaerobic, gram (+), spore forming bacillus
- Occurs in infants younger than 12 months
- Spores in Honey! No honey under 1 yr old!
- Physical findings: constipation, **progressive hypotonia**, lethargy, feeding problems, extraocular movement impairment, drooling, poor suck and gag reflex, flushed skin, NO fever, possible respiratory failure
- Toxin inhibits acetylcholine release
- Botulism Immune Globulin IV (BIG-IV), supportive care



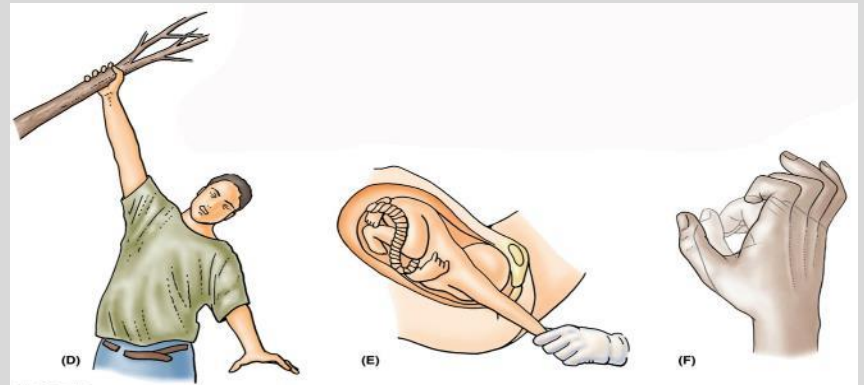
Bell Palsy

- Unilateral facial nerve palsy
- Often develops around 2 weeks after a systemic infection (**herpes simplex**, **varicella-zoster**, Epstein-Barr, **Lyme**, mumps, *Mycoplasma*, *others*)
- Treatment- **eye drops/ patch**, oral prednisone



Klumpke's Palsy- Worse prognosis

- Injury to lower trunk (C8-T1)- damage to ulnar nerve
- **Weakness of intrinsic hand muscles** (thenar, hypothenar, lumbricals, interossei)
- Claw hand deformity
- Caused by: hyperabduction injuries, cervical rib (thoracic outlet syndrome)
- +/- Horner's syndrome (miosis, ptosis, anhidrosis)

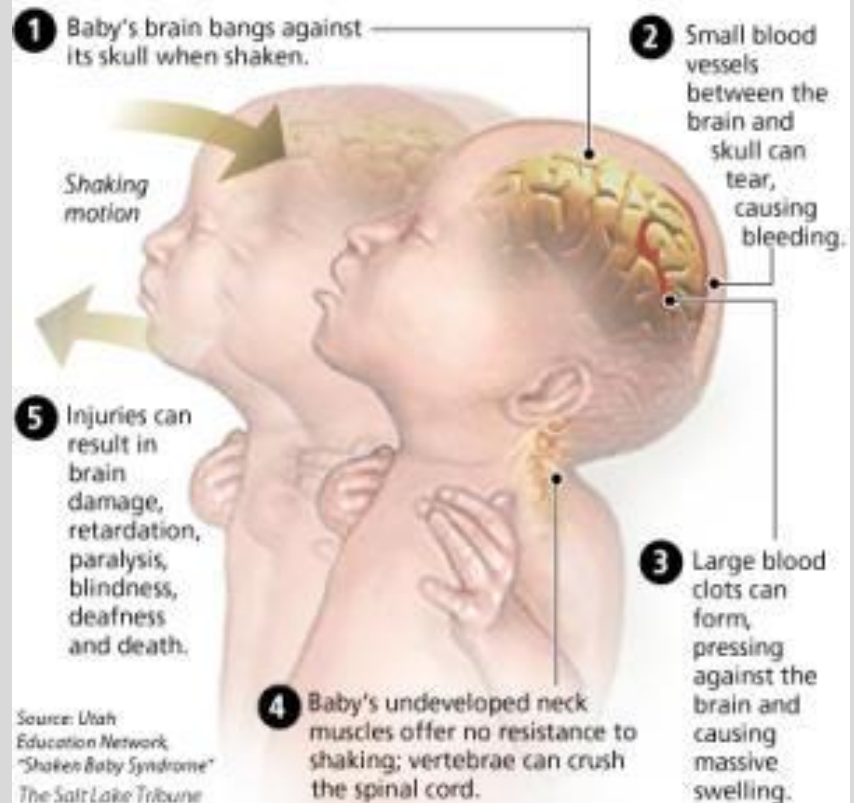


Shaken Baby Syndrome

- Intracranial injury caused by rapid acceleration and rotation of cranium
- Shearing injury of brain/intracranial vessels
- Symptoms: drowsiness, lethargy, decreased feeding, vomiting, apnea, seizures
- Physical manifestations: **subdural hematoma, retinal hemorrhage**

Damage caused when a baby is shaken

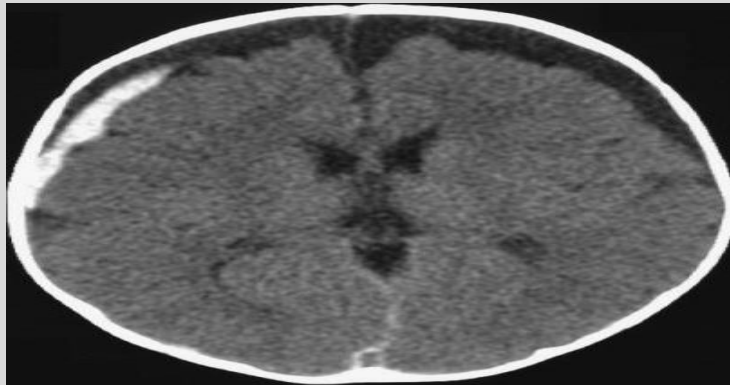
Babies are especially susceptible to injury when they are shaken because their connecting tissues and bone structure have not sufficiently developed to offer any protection.



Shaken Baby Syndrome



- Part of Anticipatory guidance
- Watch for cranky/colic infants, cranky adults, poor coping mechanisms, lack of social support



Concussion

- Traumatic forces on brain result in short-lived neurologic impairment that typically resolves spontaneously
- Immediate or evolves over minutes to hours
- Symptoms can include somatic, cognitive, emotional, physical, and sleep disturbance. Photosensitivity.
- Treatment:
 - physical rest- no work or school, no activities early on
 - cognitive rest- no reading, no Texting/TV, dark room, sunglasses, no loud music/noise, quality sleep/rest
 - Can engage in some light low intensity activities if feeling better (Prolonged rest may prolong symptoms.). Build back slowly.
 - recovery time varies.

Summary Slide 1

- Tests for hypotonia- Prone, supine, hold infant up- if slips thru fingers, then hypotonic
- Cerebral Palsy- non progressive motor impairment, upper motor neuron signs- spasticity, increased DTR's, clonus
- Spinal Muscular Atrophy (SMA) anterior horn cell, severe hypotonia, normal intelligence, fasciculations of tongue, poor feeding
- Down Syndrome- Hypotonia and other physical signs
- Prader Willi- Paternal 15q11.2-q13), infant hypotonic and then won't stop eating later in life, dev delays
- Dermatomyositis- Classic skin findings with proximal muscle weakness
- Duchenne MD- Problem with dystrophin, Weakness develops in early childhood, males, pseudohypertrophy in calves, Elevated CK, cardiomyopathy and pulmonary issues
- Becker- milder form of MD

Summary Slide Part 2

- Cephalohematoma doesn't cross suture lines,
- Caput Succedaneum does cross
- Positional plagiocephaly More belly time when awake, ROM neck exercises if torticollis, Helmet sometimes, still sleep on back
- Craniosynostosis- asymmetry to face and head, ridging of sutures, Get 3D CT scan and neurosx consult
- Meningitis- bulging fontanel, fever or hypothermia, apnea, lethargy/irritable
- Botulism- No honey under 1 yr old, hypotonia, difficulty eating, drooling, poor suck or gag reflex
- Bells palsy- protect the eye from drying out
- Erb's palsy- asymmetric moro reflex
- Klumpke palsy- loss of grip, can have a horner syndrome
- Shaken baby- subdural hematomas, retinal hemorrhages
- Concussion- Varied symptoms, need full recovery to go back to full activity